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Electrical Engineering Module

Piano Deliverable #3

Technical Memorandum Deliverable #3

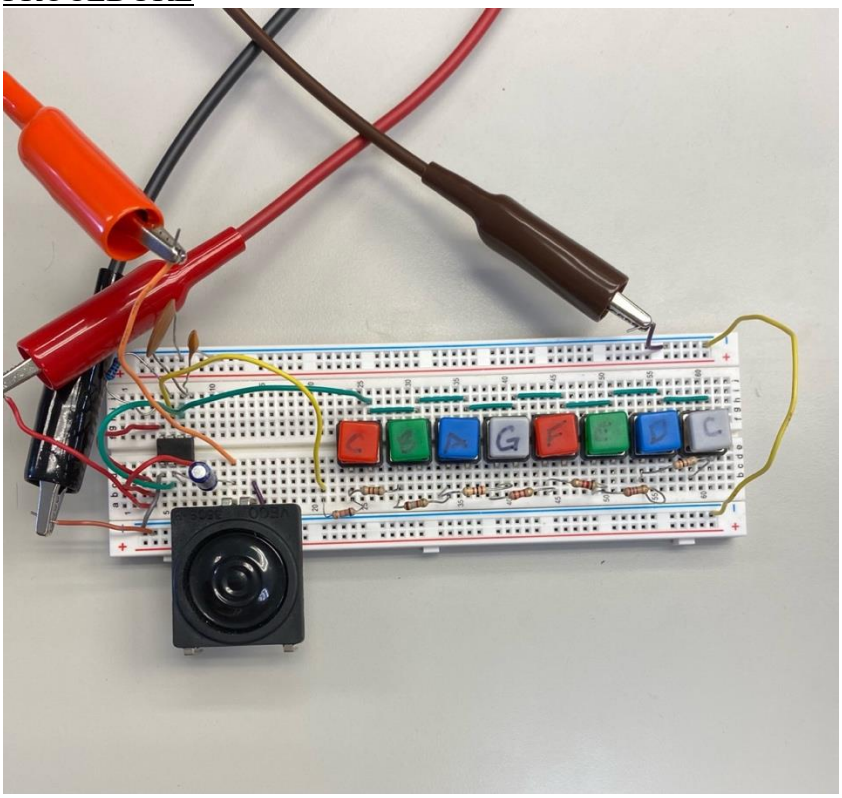
ABSTRACT

In this project we were tasked to build an electronic piano with 8 buttons, using a LM55 circuit, to generate different frequencies that play different notes. This piano was powered by 5 volts of electricity. In this memorandum we will detail our approach to this project, the steps we took to create this piano, the results and graphs that were created based on the data collected, and a reflection on our work and creation.

INTRODUCTION

In this assignment, we were instructed to build an electronic piano with 8 buttons/notes using an LM555 pulse generator. Our approach going into this project was to build the piano using the circuit breadboard example, calculating the frequencies of one octave (8 different notes), and then calculating which resistors to use in our piano to produce the desired note frequencies.

PROCEDURE



To create our circuit, we first started by using an LM555 circuit for pulse generation. We run our buttons, resistors, power, and speaker throughout the LM555 circuit. We first connected the buttons with repeating wires, so the current continues through them. We then used a frequency calculation of 1 / (.693(RA + 2RB)C) with an RA resistor of 1100 Ω to find the resistor values needed to get our desired frequencies for the various piano notes. After adding the resistors, we connected the speaker and tested the notes. Between the speaker and the LM555 circuit, we added a capacitor of 10uf to ensure the frequency going into the speaker would not blow it out. We also have two capacitors grounded from pin 5 and 6. We are running 5V of power into our circuit through pin 8 as the positive and pin 1 as the negative.

RESULTS

Explanation of Piano Set Up

Using the equation, 1 / (.693(RA + 2RB)C), we were able to calculate the different desired frequencies. When solved for RB, this equation becomes RB = 0.5*(1/.693Cf - RA). Because we wanted notes in octave 4, we analyzed the given chart and chose the 1.00x10^-7 capacitor and the 1100 Ohm resistor. When we insert those two constants and the desired RA frequency for the first note, C, we were able to solve for the RB resistor we needed to use to generate the desired note. We chose the 13,000 Ohm resistor first based on the RB value of 13,239, and then we subtracted the subsequent RB values from their previous values to choose the other resistors.

Frequency Calculations

Insert C	C =	1.00E-07	F
Insert R_A	R_A =	1100	Ω

Insert Note	***Insert Freq***	Calculated R_B
C	261.63	27027
D	293.66	24019
E	329.63	21338
F	349.23	20110
G	392	17856
A	440	15848
B	493.88	14059
C	523.25	13239

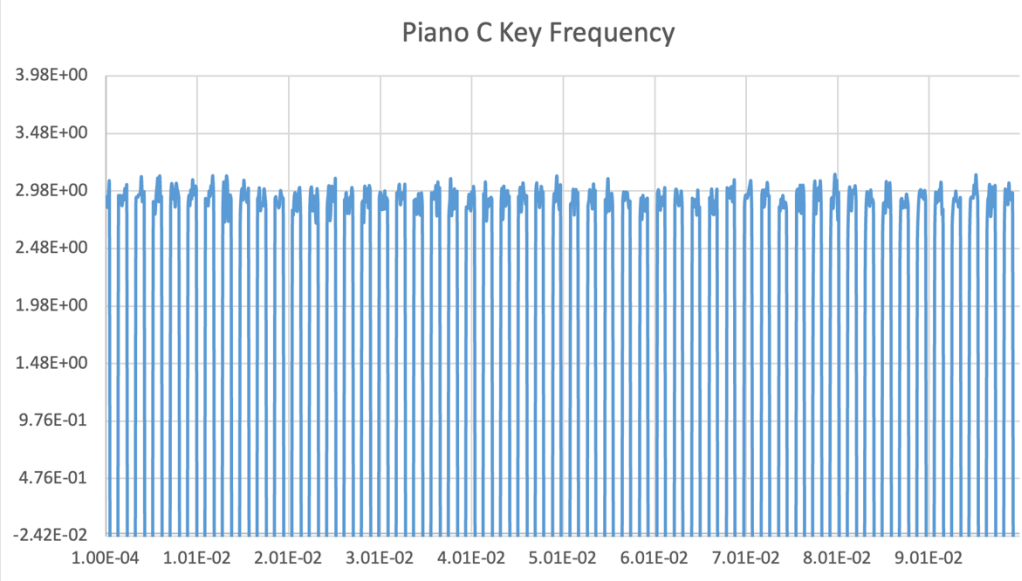
Pictures of Measured Frequencies from Multimeter

C	B	A	G
			
F	**E**	**D**	**C**
			

Waveform of Time and Frequency Domain from the Oscilloscope (Key C)



Time Domain Plotted in Excel (Key C)



X axis: Time milli-seconds
Y axis: Frequency

CONCLUSION

In this project, we learned how to generate different frequencies using an LM55 pulse generator. We learned about frequencies, resistance, and duty cycles. Specifically, we learned how to use the oscilloscope to measure waveform and how to use a digital multimeter to measure frequency. We applied basic circuitry skills to build, and problem shoot the electronic piano. Now, knowing better how circuits must be grounded and how they work in general, we could have improved our efficiency in building the circuit and generating the frequencies. We most enjoyed learning how to put the circuit together and build the piano.